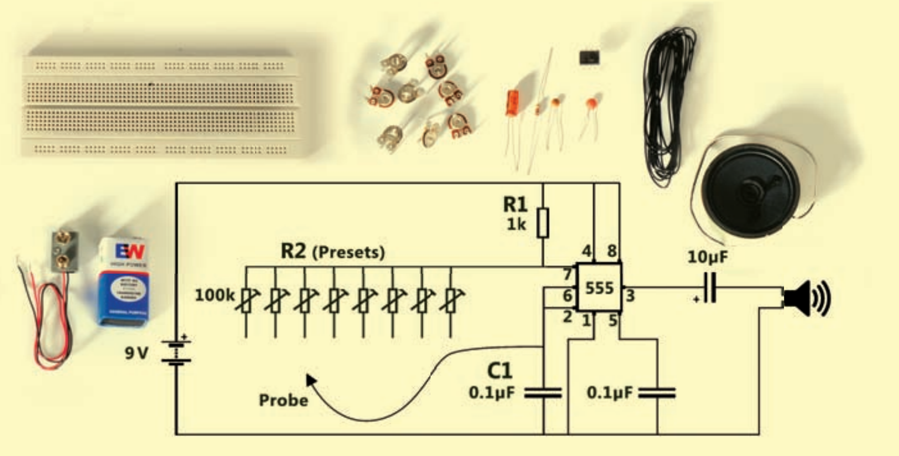
Make some melody

Although playing a piano can be fun (even if you don't know music), nothing comes close to the joy of playing a piano you built yourself!



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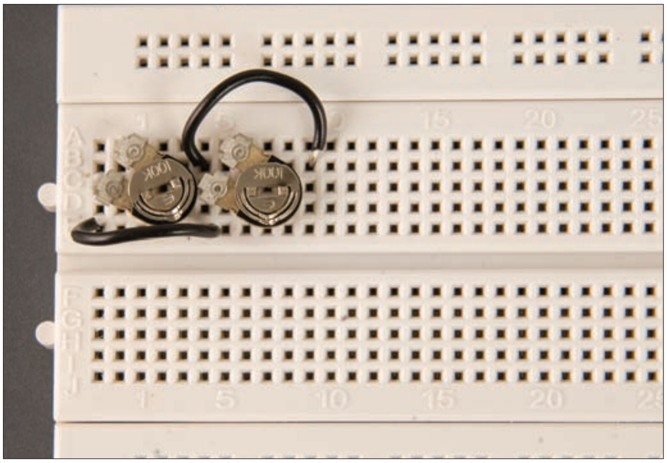
A piano is an old instrument that uses keys to cause hammers to strike tuned strings. Today the term piano is popularly used (rather ignorantly) for most musical instruments with keys on them! The ideal term would be synthesizer because what they primarily do is synthesize or recreate the sound of a musical instrument electronically.

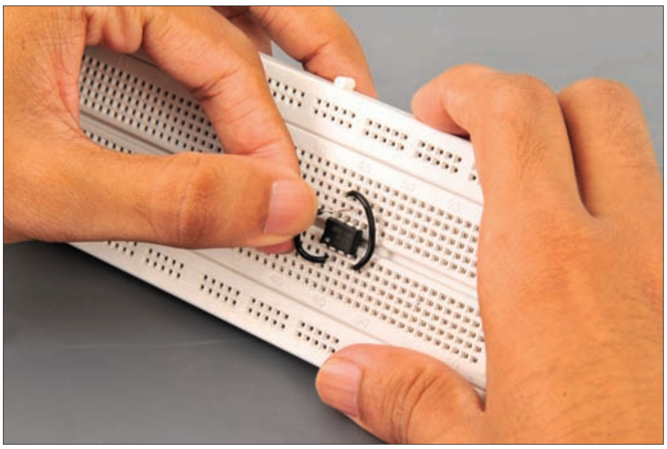
In order to understand how a basic synthesizer works, we need to understand what goes behind it. Music to be precise. Those sheets musicians read from while performing are made up of octaves or combinations of 8 basic notes from A to G. Depending upon the medium of music, be it Western or Indian, they are expressed as Do, Re, Me, Fa, So, La, Ti and Do; or Sa, Re, Ga, Ma, Pa, Dha, Ni, and Sa. Essentially, these have distinct frequencies, and all instruments are tuned according to a tuner emitting the specific frequencies. In order to create our basic single-tone, single-octave synthesizer, we'll require the frequencies in a single octave. Considering note A as the base, the rest follow an increase and decrease in frequencies with respect to A.

Key	Notation	Hz
63	B5	987.77
61	A5	880
59	G5	783.99
57	F5	698.46
56	E5	659.26
54	D5	587.33
52	C5	523.25
51	B4	493.88
49	A4 concert pitch	440
47	G4	392
45	F4	349.23
44	E4	329.63
42	D4	293.67
40	C4	261.63

The notes differ in frequencies and these sets are replicated for additional octaves



Each preset will decide the pitch and note of the octave



The pin near the notch is pin 1. Here pin 4 (bottom right) is connected to pin 8, and pin 2 is being connected to pin 6

What you need

IC 555, 9V battery, preset 100k – 8 nos, 0.1 uF ceramic capacitors – 2 nos, 1k resistor – 1 no., 10 uF electrolytic capacitor – 1 no, speaker – 1 no, connecting wire, bread board – 1 no., multimeter 9V battery snap – 1 no, tiny screwdriver for tuning frequencies.

Once you are able to generate these frequencies, you have a functional synthesizer. Our "pocket piano" as we'll call it, is based on a 555 timer IC. This is a really cheap and multi-functional chip used in timer and counting circuitries.

The configuration of the circuit is called an astable multivibrator. Such a circuit does not need any input trigger, but rather oscillates into desired frequency depending upon the

How to use a breadboard

In a typical breadboard as shown in the image, the grid in the middle is connected vertically in sets of 5 holes each. Similarly, the grid on the top and bottom of the breadboard is connected horizontally. If you have to connect ICs where there are multiple pins that should be connected together, use the gap in the middle of the breadboard. This way the pins on either side are isolated. Using connecting wires you can make the required connections.